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Mobile Communication System

Wireless Communication

- Wireless communication is fastest growing technology.
- A method of transmitting information from one point to the other without using any transmission medium.

Why wireless communication?

- Mobility, flexibility and ease of use.
- Ease of installation and reliability.
- Disadvantages include interference and security.

Wireless Communication

- Basic Elements
  - Transmitter
  - Channel(space)
  - Receiver
- Some important wireless communication system include
  - WLAN (Wi-Fi)
  - Bluetooth
  - Mobile Telephone System (Cellular Communication)
  - Television and Radio Broadcasting
  - Satellite Communication

Wireless LAN

- IEEE 802.11, is the specification for wireless LAN.
- Architecture
  - Basic Service Set(BSS)
  - Extended Service Set(ESS)

802.11 Architecture

- Basic Service Set
  - The BSS without an AP(central base station) is a stand-alone network and cannot send data to other BSSs.
  - It is called an ad hoc architecture.
  - BSS with an AP is called infrastructure network.

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802.11 Architecture

- Extended Service Set
  - Two or more BSS with Aps.
  - Distribution System connects the BSSs and Aps.
  - Two types of stations, **Mobile and Stationary stations**.

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MAC Sublayer

- IEEE 802.11 defines two MAC sublayers:
  - Distributed Coordination Function (DCF).
  - Point Coordination Function (PCF).

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Distributed Coordination Function

- DCF uses CSMA/CA as the access method.

Sending a Frame

- Before sending a frame source senses the medium.
- After the channel is found to be idle, the station waits for a period of time called the distributed interframe space (DIFS); then the station sends a control frame called the request to send (RTS).
- After receiving the RTS and waiting a period of time called the short interframe space (SIFS), the destination station sends a control frame, called the clear to send (CTS), to the source station.
- The source station sends data after waiting an amount of time equal to SIFS.
- The destination station, after waiting an amount of time equal to SIFS, sends an acknowledgment to show that the frame has been received.

Network Allocation Vector

- Collision avoidance feature.
- RTS frame includes the duration of time that it needs to occupy the channel .
- A network allocation vector (NAV) that shows how much time must pass before these stations are allowed to check the channel for idleness.
- Each time a station accesses the system and sends an RTS frame, other stations start their NAV.
- Each station, before sensing the physical medium to see if it is idle, first checks its NAV to see if it has expired.

Bluetooth

- Bluetooth is a wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers, cameras, printers, coffee makers, and so on.
- A Bluetooth LAN is an ad hoc network, which means that the network is formed spontaneously.
- Gadgets find each other and make a network called a piconet.
- Bluetooth technology is the implementation of a protocol defined by the IEEE 802.15 standard.

Architecture

- Bluetooth defines two types of networks:
  - Piconet
  - Scatternet

Piconet

- A piconet can have up to eight stations, one primary and others secondary.
- All the secondary stations synchronize their clocks and hopping sequence with the primary.
- The communication between the primary and the secondary can be one-to-one or one-to-many.
- Although a piconet can have a maximum of seven secondaries, an additional eight secondaries can be in the parked state.

Scatternet

- Piconets can be combined to form called a scatternet.
- A secondary station in one piconet can be the primary in another piconet.
- This station can receive messages from the primary in the first piconet and, acting as a primary, deliver them to secondaries in the second piconet.
- A station can be a member of two piconets.

Bluetooth Layers

- Radio Layer
  - The radio layer is roughly equivalent to the physical layer of the Internet model.
  - Bluetooth devices are low-power and have a range of 10 m.
- Baseband Layer
  - The baseband layer is roughly equivalent to the MAC sublayer in LANs.
  - The access method is TDMA.
  - The primary and secondary communicate with each other using time slots.

Bluetooth Layers

- L2CAP
  - Equivalent to LLC sublayer in LAN.
  - It is used for data exchange on an ACL link.
  - The L2CAP has specific duties: multiplexing, segmentation and reassembly, quality of service (QoS), and group management.
- Other Upper Layers
  - Bluetooth defines several protocols for the upper layers that use the services of L2CAP; these protocols are specific for each purpose.

Cellular Telephony

- To provide communications between two mobile stations (MSs), or between one mobile unit and one stationary unit, often called a land unit.
- each cellular service area is divided into small regions called cells.
- Each cell contains an antenna and is controlled by a solar or AC powered network station, called the base station (BS).
- Each base station is controlled by a switching office, called a mobile switching center (MSC).
- The MSC coordinates communication between all the base stations and the telephone central office.

Cellular Network Components

- BTS (Base Transceiver Station) – main component of a cell and it connects the subscribers to the cellular network; for transmission/reception of information it uses several antennas spread across the cell
- BSC (Basic Station Controller) – it is an interface between BTSs and it is linked to BTSs by cable or microwave links; it routes calls between BTSs; it is also connected to the MSC
- MSC (Mobile Switching Center) – the coordinator of a cellular network, it is connected to several BSCs, it routes calls between BSCs; links the cellular network with other networks like PSTN through fiber optics, microwave or copper cable

Features cellular concept

Frequency Reuse

- It is the technique of using same radio frequencies on radio transmitter sites within a geographic area that separated by sufficient distance to cause minimal interference with each other.

Cell Splitting

- Cell divided into smaller cells known as micro cells.

Mobile Communication

- A wireless form of communication in which voice and data information is emitted, transmitted and received via microwaves.
- A cellular phone is a portable telephone that does not use a wired connection.
- It connects to a wireless carrier network using radio waves.
- The mobile phone or cell phone is a long-range, portable electronic device used for mobile communication.
- Most current mobile phones connect to a cellular network of base stations (cell sites), which is in turn interconnected to the public switched telephone network (PSTN).

Mobile Communication

- In mobile communication technology data, voice and multimedia are transferring without having to be connected to a fixed physical media.
- The Different mobile generations:
  - 1G,2G,3G,4G,5G

Mobile Communication

- Cell phone of a user cell has its own base Transceiver station at its center.
- Whenever a call is setup the first signal is sent to the base transceiver station of the cell from this base transceiver station.
- It goes to the controller which controls the working of all the base stations from BSC.

Mobile Communication

- It then goes to MSC or mobile switching center which is master controller of the entire system.
- These MSC are different for different areas. From MSC of area one the signal is transmitted to MSC of area two.
- Where it follows the reverse sequence as MSC to BSC, BSC to BTS and from BTS-MS.

Mobile Communication System

- Whenever a user tries to call someone.
- A separate channel assigned to the user by the MSC.
- If all the channels already occupied by other user, then its user must wait for a channel to become free.
- In such a case the user gets the notification such as call cannot be completed or network error.